

Nuclear density

$$\rho = \frac{\text{mass of the nuclei}}{\text{Volume of the nuclei}} = \frac{A.m}{\frac{4}{3}\pi R_0^3 A} = \frac{m}{\frac{4}{3}\pi R_0^3}$$

The above expression shows that the nuclear density is independent of the mass number A. In other words, all the nuclei ($Z > 10$) have the same density and it is an important characteristic property of all nuclei.

We can calculate the numerical value of this density by substituting the corresponding values.

$$\rho = \frac{1.67\times10^{-27}}{\frac{4}{3}\pi\times(1.2\times10^{-15})^3} = 2.3\times10^{17}\,\text{kg m}^{-3}.$$

It implies that nucleons are extremely tightly packed or compressed state in the nucleus and compare this density with the density of water which is $10^3\,\text{kg m}^{-3}$.



A single teaspoon of nuclear matter would weigh about trillion tons.

9.4.5 Mass defect and binding energy

It is experimentally found out that the mass of any nucleus is always less than the sum of the masses of its individual constituent particles. For example, consider the carbon-12 nucleus which is made up of 6 protons and 6 neutrons.

Mass of 6 neutrons $= 6\times1.00866\ u = 6.05196\,u$

Mass of 6 protons $= 6\times1.00727\,u = 6.04362\,u$

Mass of 6 electrons $= 6\times0.00055\,u = 0.0033\,u$

The expected mass of carbon-12 nucleus $= 6.05196\,u + 6.04362\,u = 12.09558\,u$

But using mass spectroscopy, the atomic mass of carbon-12 atom is found to be 12 u. So if we subtract the mass of 6 electrons (0.0033 u) from 12 u, we get the nuclear mass of carbon-12 atom which is equal to 11.9967 u. Hence the experimental mass of carbon-12 nucleus is less than the total mass of its individual constituents by $\Delta m = 0.09888\,u$. This difference in mass Δm is called mass defect. In general, if M, m_p, and m_n are mass of the nucleus (^A_ZX), the mass of a proton and the mass of a neutron respectively, then the mass defect is given by

$$\Delta m = \left(Zm_p + Nm_n\right) - M \qquad (9.20)$$

Where has this mass disappeared? The answer was provided by Albert Einstein with the help of famous mass-energy relation ($E = mc^2$). According to this relation, the mass can be converted into energy and energy can be converted into mass. In the case of the carbon-12 nucleus, when 6 protons and 6 neutrons combine to form carbon-12 nucleus, mass equal to mass defect disappears and an energy equivalent to missing mass. This energy is called the binding energy of the nucleus (BE) and is equal to $(\Delta m)c^2$. In fact, to separate the carbon-12 nucleus into individual constituents, we must supply the energy equal to binding energy of the nucleus.

We can write the equation (9.20) in terms of binding energy

$$BE = \left(Zm_p + Nm_n - M\right)c^2 \qquad (9.21)$$

It is always convenient to work with the mass of the atom rather than with the mass of

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